

Applied Physics

Chapter no: 3

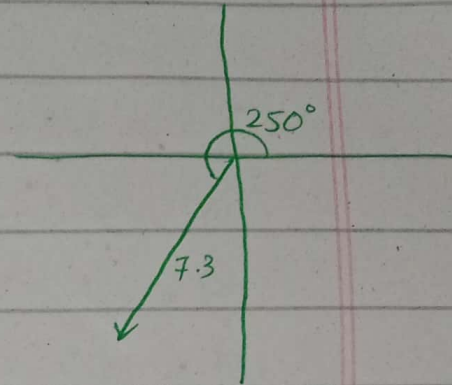
Q no: 1

Given data

$$\theta = 250^\circ$$

$$d = 7.3 \text{ m}$$

$$d_x = ? \quad ; \quad d_y = ?$$



Solution:

$$d_y = d \sin \theta \quad ; \quad d_x = d \cos \theta$$

$$d_y = 7.3 \sin 250 \quad ; \quad d_x = 7.3 \cos 250$$

$$d_y = 6.85 \text{ m} \quad ; \quad d_x = 2.49 \text{ m}$$

Q no: 5

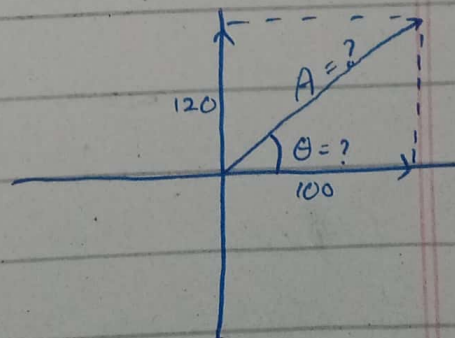
Given data

$$A_x = 100 \text{ km}$$

$$A_y = 120 \text{ km}$$

$$A = ?$$

$$\theta = ?$$



Solution:

$$A = \sqrt{A_x^2 + A_y^2}$$

$$A = \sqrt{(100)^2 + (120)^2}$$

$$A = 156.20 \text{ km}$$

$$\theta = \tan^{-1} \frac{A_y}{A_x}$$

$$\theta = \tan^{-1} \left(\frac{120}{100} \right)$$

$$\theta = 50.19^\circ$$

Q no: 6

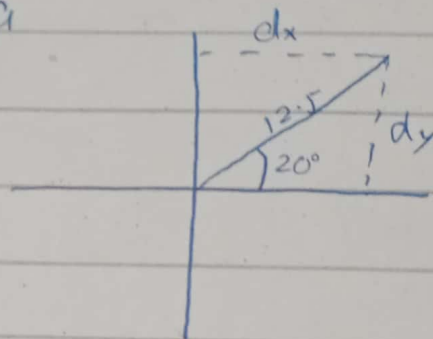
Given Data

$$\theta = 20^\circ$$

$$d = 12.5 \text{ m}$$

$$d_x = ?$$

$$d_y = ?$$



Solution:

$$d_x = d \cos \theta \quad ; \quad d_y = d \sin \theta$$

$$d_x = 12.5 \cos 20^\circ \quad ; \quad d_y = 12.5 \sin 20^\circ$$

$$d_x = 11.74 \text{ m} \quad ; \quad d_y = 4.27 \text{ m}$$

Q no: 8

Given data

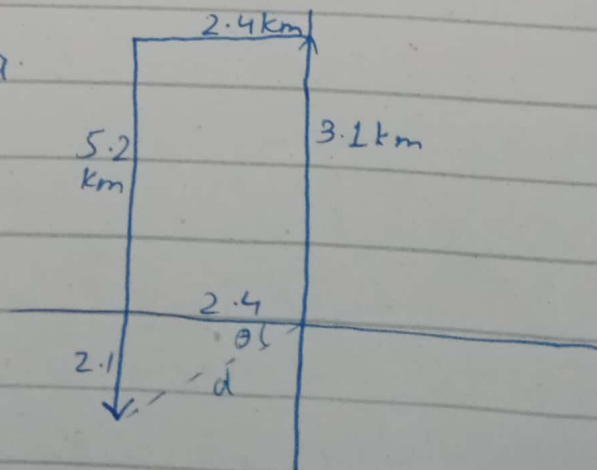
$$d_x = 2.4 \text{ km}$$

$$d_y = 5.2 - 3.1$$

$$d_y = 2.1 \text{ km}$$

$$d = ?$$

$$\theta = ?$$



Solution:

$$d = \sqrt{dx^2 + dy^2}$$

$$d = \sqrt{(2.4)^2 + (2.1)^2}$$

$$d = 3.18$$

$$\phi = \frac{dy}{dx} = \frac{2.1}{2.4} = \theta = \tan^{-1} \frac{dy}{dx}$$

$$\theta = \tan^{-1} \left(\frac{2.1}{2.4} \right)$$

$\theta = 41.18^\circ$ with -ve x-axis

221.18 with +ve x-axis

Q no: 13

Given data

$$d = 3.4 \text{ km}$$

$$\theta = 35^\circ$$

$$dy = ? ; dx = ?$$

$$\text{Total distance} = dy + dx$$

Solution:

$$dy = d \sin \theta ; dx = d \cos \theta$$

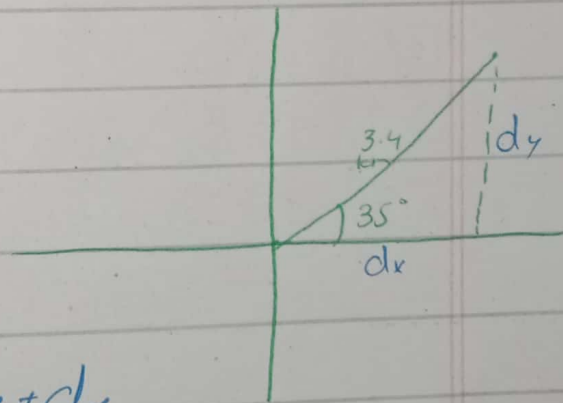
$$= 3.4 \sin 35 ; = 3.4 \cos 35$$

$$= 1.95 \text{ km} ; = 2.78 \text{ km}$$

$$\text{Total / shortest distance} = dx + dy$$

$$= 1.95 + 2.78$$

$$= 4.73 \text{ km.}$$



Q no: 15

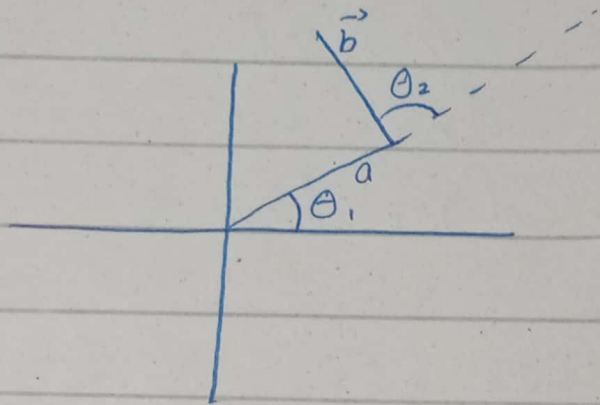
Given data

$$|a| = |b| = 10 \text{ cm}$$

$$\theta_1 = 30^\circ ; \theta_2 = 105^\circ$$

$$|r| = ? ; r_x = ? ; r_y = ?$$

$$\theta = ?$$



Solution:-

$$r = \vec{a} + \vec{b}$$

For $\vec{a}$

$$\vec{a} = a \cos \theta_1 \hat{i} + a \sin \theta_1 \hat{j}$$

$$\vec{a} = 8.66 \hat{i} + 5 \hat{j}$$

For $\vec{b}$

$$\vec{b} = b \cos(\theta_1 + \theta_2) \hat{i} + b \sin(\theta_1 + \theta_2) \hat{j}$$

$$\vec{b} = -7.07 \hat{i} + 7.07 \hat{j}$$

r_x

$$r_x = a \cos \theta_1 + b \cos(\theta_1 + \theta_2)$$

$$= 8.66 - 7.07$$

$$= 1.59 \hat{i}$$

r_y

$$\begin{aligned} r_y &= a \sin \theta_1 + b \sin(\theta_1 + \theta_2) \hat{j} \\ &= 5 + 7.07 \\ &= 12.07 \hat{j} \end{aligned}$$

$\vec{r}$

$$\begin{aligned} \vec{r} &= r_x + r_y \\ &= 1.59 \hat{i} + 12.07 \hat{j} \end{aligned}$$

$|r|$

$$|r| = \sqrt{(1.59)^2 + (12.07)^2}$$

$$|r| = 12.17 \text{ m}$$

θ

$$\theta = \tan^{-1} \left(\frac{r_y}{r_x} \right) = \tan^{-1} \left(\frac{12.07}{1.59} \right)$$

$$\theta = 82.5^\circ \text{ with +ve x-axis.}$$

Q no : 20

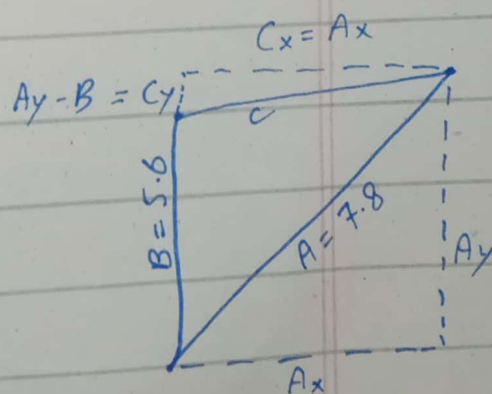
Given data

$$A = 7.81 \text{ km with } 50^\circ$$

$$B = 5.6 \text{ km}$$

$$C = ?$$

$$\theta = ?$$



Solution :-

$$A_x = A \cos \theta$$

$$A_x = 7.8 \cos 50^\circ$$

$$A_x = 5.01 \text{ km}$$

$$A_y = A \sin \theta$$

$$= 7.8 \sin 50^\circ$$

$$= 5.97 \text{ km}$$

$$C_y = A_y - B$$

$$= 5.97 - 5.6$$

$$= 0.375 \text{ km}$$

$$C = \sqrt{C_x^2 + C_y^2}$$

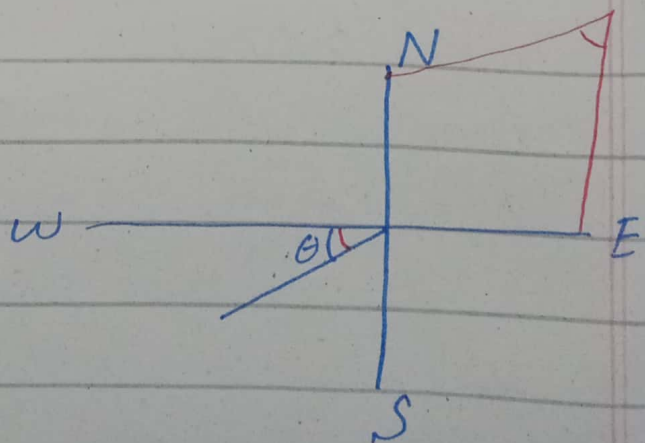
$$C = \sqrt{(5.01)^2 + (0.375)^2}$$

$$C = 5.02 \text{ km}$$

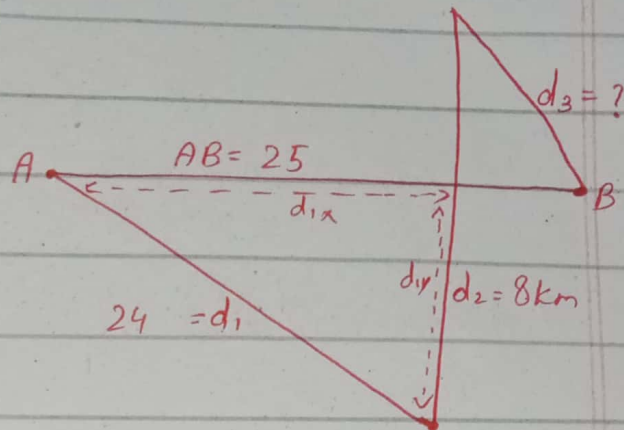
for direction

$$\theta = \tan^{-1} \frac{C_y}{C_x} = \tan^{-1} \left(\frac{0.375}{5.02} \right)$$

$\theta = 4.28^\circ$ in south of due west



Q: no: 25:



Given data

$$AB = 25 \text{ km}$$

$$d_1 = 24 \text{ km with } 15^\circ$$

$$d_2 = 8 \text{ km}$$

$$d_3 = ?$$

Solution:-

$$d_{1x} = 24 \cos 15 = 23.18 \text{ km}$$

$$d_{1y} = 24 \sin 15 = 6.21 \text{ km}$$

$$\begin{aligned} d_{3x} &= AB - d_{1x} \\ &= 25 - 23.18 \\ &= 1.82 \text{ km} \end{aligned}$$

$$\begin{aligned} d_{3y} &= d_2 - d_{1y} \\ &= ~~8.4~~ 8 - 6.21 \\ &= 1.79 \text{ km} \end{aligned}$$

$$\begin{aligned}
 d_3 &= \sqrt{d_{3x}^2 + d_{3y}^2} \\
 &= \sqrt{(1.82)^2 + (1.79)^2} \\
 &= 2.55 \text{ km}
 \end{aligned}$$

Q no : 27

Given data

$$\vec{d}_1 + \vec{d}_2 = 5\vec{d}_3 \quad - \text{eq (i)}$$

$$\vec{d}_1 - \vec{d}_2 = 3\vec{d}_3 \quad - \text{eq (ii)}$$

$$\vec{d}_3 = 2\hat{i} + 4\hat{j} \quad - \text{eq (iii)}$$

Solution:-

Adding eq (i) and eq (ii)

$$\vec{d}_1 + \vec{d}_2 = 5\vec{d}_3$$

$$\vec{d}_1 - \vec{d}_2 = 3\vec{d}_3$$

$$2\vec{d}_1 = 8\vec{d}_3$$

$$\vec{d}_1 = 4\vec{d}_3 \quad - \text{eq (iv)}$$

So,

$$\vec{d}_1 = 4(2\hat{i} + 4\hat{j})$$

$$\vec{d}_1 = 8\hat{i} + 16\hat{j}$$

Using 2nd eq (ii)

$$\vec{d}_1 - \vec{d}_2 = 3\vec{d}_3$$

$$\vec{d}_2 = \vec{d}_1 - 3\vec{d}_3$$

by eq (iv) $\vec{d}_1 = 4\vec{d}_3$

$$\vec{d}_2 = 4\vec{d}_3 - 3\vec{d}_3$$

$$\vec{d}_2 = \vec{d}_3$$

$$\vec{d}_2 = 2\hat{i} + 4\hat{j}$$

Q no: 33

Given data

$$a = 4$$

$$b = 3$$

$$c = 5$$

a. $|\vec{a} \times \vec{b}| = ?$

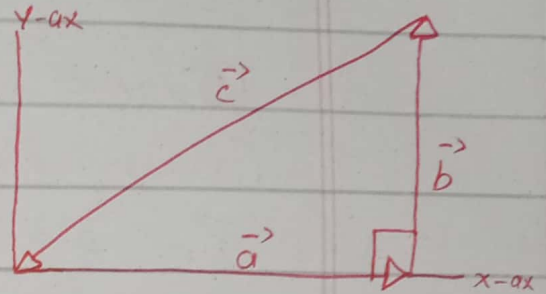
b. direction = ?

c. $|\vec{a} \times \vec{c}| = ?$

d. direction = ?

e. $|\vec{b} \times \vec{c}| = ?$

f. direction = ?



Solution:-

a) $\vec{a} \times \vec{b} = a.b \sin \theta$
 $= 4.3 \sin 90^\circ$
 $= 12$

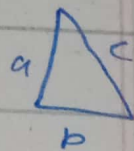
b) +ve z - axis.

c) $\vec{a} \times \vec{c} = ac \sin \theta$
 $= ac \left(\frac{b}{c} \right) \therefore \sin \theta = (b/c)^{P/H}$
 $= ab = 12$

d) direction is in -ve z-axis.

$$e) \vec{b} \times \vec{c} = bc \sin \theta$$

$$= bc \frac{a}{c}$$



$$= ab = 12$$

f) direction is in +z-axis.

③ no: 37.

Given data

$$\vec{a} = 3\hat{i} + 3\hat{j} - 2\hat{k}$$

$$\vec{b} = -1\hat{i} - 4\hat{j} + 2\hat{k}$$

$$\vec{c} = 2\hat{i} + 2\hat{j} + 1\hat{k}$$

Find Out:

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = ?$$

$$\vec{a} \cdot (\vec{b} + \vec{c}) = ?$$

$$\vec{a} \times (\vec{b} + \vec{c}) = ?$$

Solution:

$$a) \vec{b} \times \vec{c} = \begin{vmatrix} -1 & -4 & 2 \\ 2 & 2 & 1 \end{vmatrix}$$

$$\vec{b} \times \vec{c} = (-4 - 4)\hat{i} - (-1 - 4)\hat{j} + (-2 + 8)\hat{k}$$

$$= -8\hat{i} + 5\hat{j} + 6\hat{k}$$

$$\begin{aligned}\vec{a} \cdot (\vec{b} \times \vec{c}) &= (3\hat{i} + 3\hat{j} - 2\hat{k}) \cdot (-8\hat{i} + 5\hat{j} + 6\hat{k}) \\ &= -24 + 15 - 12 \\ &= -21\end{aligned}$$

$$\begin{aligned}(b) \quad (\vec{b} + \vec{c}) &= -1\hat{i} - 4\hat{j} + 2\hat{k} \\ &\quad 2\hat{i} + 2\hat{j} + 1\hat{k} \\ (\vec{b} + \vec{c}) &= 1\hat{i} - 2\hat{j} + 3\hat{k}\end{aligned}$$

$$\begin{aligned}\vec{a} \cdot (\vec{b} + \vec{c}) &= (3\hat{i} + 3\hat{j} - 2\hat{k}) \cdot (1\hat{i} - 2\hat{j} + 3\hat{k}) \\ &= 3 - 6 - 6 \\ &= -9\end{aligned}$$

$$(c) \quad \vec{a} \times (\vec{b} + \vec{c}) = \begin{vmatrix} 3 & 3 & -2 \\ 1 & -2 & 3 \end{vmatrix}$$

$$\begin{aligned}\vec{a} \times (\vec{b} + \vec{c}) &= (9 - 4)\hat{i} - (9 + 2)\hat{j} + (-6 - 3)\hat{k} \\ &= 5\hat{i} - 11\hat{j} - 9\hat{k}\end{aligned}$$

Q no : 39 :

Given Data

$$|A| = 6$$

$$|B| = 7$$

$$\vec{A} \cdot \vec{B} = 14$$

$$\theta = ?$$

Solution:

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$\theta = \cos^{-1} \frac{\vec{A} \cdot \vec{B}}{AB}$$

$$\theta = \cos^{-1} \left(\frac{14}{6 \times 7} \right)$$

$$\theta = \cancel{1.2309}$$

$$70.5^\circ$$

Question no: 43

Given Data

$$|a| = 3 \quad a_x = ? \quad a_y = ?$$

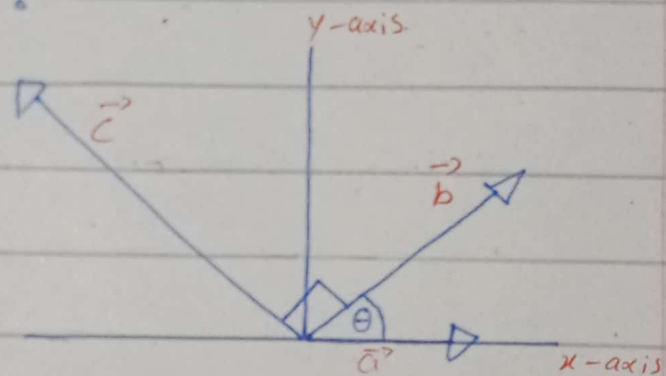
$$|b| = 4 \quad b_x = ? \quad b_y = ?$$

$$C = 10 \quad C_x = ? \quad C_y = ?$$

$$\theta = 30^\circ$$

$$\vec{C} = p\vec{a} + q\vec{b}$$

$$p = ? \quad ; \quad q = ?$$



Solution:

$$a_x = a \cos \theta = 3 \cos 0 = 3\text{m}$$

$$a_y = a \sin \theta = 3 \sin 0 = 0$$

$$b_x = b \cos \theta = 4 \cos 30 = 3.46\text{m}$$

$$b_y = b \sin \theta = 4 \sin 30 = 2\text{m}$$

ex: As seen from graph

$$\theta_2 = 90, \text{ so } \theta_c = \theta_1 + \theta_2 = 120^\circ$$

$$C_x = \cancel{C \sin} C \cos \theta_c = 10 \cos 120 = -5\text{m}$$

$$C_y = C \sin \theta_c = 10 \sin 120 = \cancel{8.66} 8.66\text{m}$$

$$\vec{c} = p\vec{a} + q\vec{b}$$

put x and y intercepts, separately

$$-5 = 3p + 3.46q \quad \text{--- (i)}$$

$$8.66 = 0p + 2q \quad \text{--- (ii)}$$

From (ii)

$$q = 8.66/2 = 4.33m$$

put q in eq(i)

$$-5 = 3p + 3.46(4.33)$$

$$3p = -5 - 14.98$$

$$p = -19.98/3$$

$$p = -6.6606$$

Check points & Examples

NO:1

$$|\vec{a}| = 3m$$

$$|\vec{b}| = 4m$$

$$\vec{c} = \vec{a} + \vec{b}$$

On basis of orientation

$$\text{Max} = |\vec{c}| = 7$$

$$\text{Min} = |\vec{c}| = -1 (\text{or } 1)$$

Check-point-no-2:

c, d, and f are proper to determine vector.

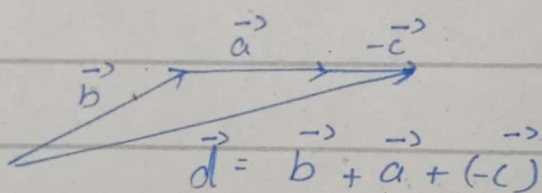
Example no:1

$\vec{a} = 2\text{km}$ via east

$\vec{b} = 2\text{km}$ north of east 30°

$\vec{c} = 1\text{km}$ via west

longest distance = ?



$$\vec{d} = \vec{b} + \vec{a} + (-\vec{c})$$

$$\vec{d} = 4.8\text{km}$$

Example no:2

$$\vec{d} = 215\text{km}$$

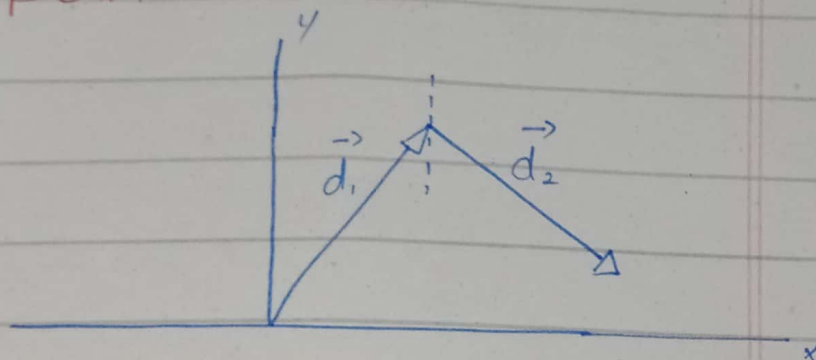
22° is with ~~east~~ ^{east of} north

$$\theta = (90 - 22) = 68^\circ \text{ it is north of east.}$$

$$dy = d \sin \theta = 215 \sin 68 = 199\text{km}$$

$$dx = d \cos \theta = 215 \cos 68 = 81\text{km}$$

Check-point-no-3:



- > signs of both x and y components of d_1 will be +ve.
- > for d_2
 - x-component is +ve
 - y-component is -ve
- > for $d_1 + d_2$
 - both x & y comp. are +ve.

Example-no-3.3

$$d_1 = 6$$

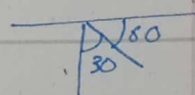
$$\theta_1 = 40$$

$$d_2 = 8$$

$$\theta_2 = 30$$

$$d_3 = 5$$

$$\theta_3 = 0$$



its with -ve y so, $\theta = -60$

$$|d_{net}| = ?$$

$$\theta_{net} = ?$$

$$d_{xnet} = d_1 \cos \theta_1 + d_2 \cos \theta_2 + d_3 \cos \theta_3 = 13.6m$$

$$d_{ynet} = d_1 \sin \theta_1 + d_2 \sin \theta_2 + d_3 \sin \theta_3 = -3.07m$$

$$d_{net} = \sqrt{d_{xnet}^2 + d_{ynet}^2}$$

$$d_{net} = 13.9m$$

$$\theta = \tan^{-1} \left(\frac{d_{ynet}}{d_{xnet}} \right)$$

$$\theta = -12.7^\circ$$

Sample Example 3.4

$$\vec{a} = 4.2 \hat{i} - 1.5 \hat{j}$$

$$\vec{b} = -1.6 \hat{i} + 2.9 \hat{j}$$

$$\vec{c} = 0 \hat{i} - 3.7 \hat{j}$$

$$r_x = a_x + b_x + c_x = 2.6 \hat{i} \text{ m}$$

$$r_y = a_y + b_y + c_y = -2.3 \hat{j} \text{ m}$$

$$\vec{r} = 2.6 \hat{i} - 2.3 \hat{j}$$

$$|r| = \sqrt{(2.6)^2 + (-2.3)^2}$$

$$|r| = 3.5 \text{ m}$$

$$\theta = \tan^{-1} \frac{r_y}{r_x} = \tan^{-1} \left(\frac{-2.3}{2.6} \right)$$

$$\theta = -41^\circ$$

Check-point 4:

$$|C| = 3, |D| = 4$$

$$C \cdot D = 0 \text{ if } \theta = 90^\circ$$

$$C \cdot D = 12 \text{ if } \theta = 0^\circ$$

$$C \cdot D = -12 \text{ if } \theta = 180^\circ$$

Check-point 5

$$|C| = 3, \quad |D| = 4$$
$$\vec{C} \times \vec{D} = 0 \quad \text{if } \theta = 0^\circ$$
$$\vec{C} \times \vec{D} = 12 \quad \text{if } \theta = 90^\circ$$

Example no: 5

$$\vec{a} = 3\hat{i} - 4\hat{j}$$

$$\vec{b} = -2\hat{i} + 3\hat{k}$$

$$\theta = ?$$

Solution

$$|a| = 5, \quad |b| = 3.61$$

$$\vec{a} \cdot \vec{b} = -6$$

$$\vec{a} \cdot \vec{b} = ab \cos \theta$$

$$\theta = \cos^{-1} \left(\frac{\vec{a} \cdot \vec{b}}{ab} \right)$$

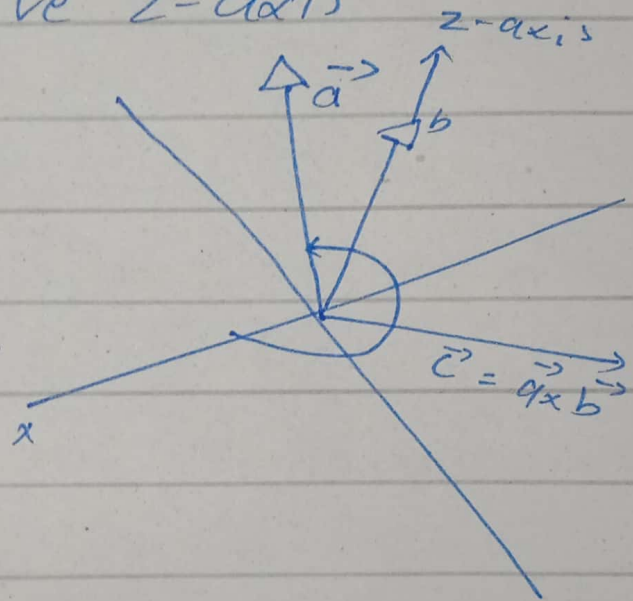
$$\theta = \cos^{-1} \left(\frac{-6}{(5 \times 3.61)} \right) = \cos^{-1} \left(-\frac{1}{3} \right)$$

$$\theta = 109.47^\circ \approx 110^\circ$$

Example no-6:

$\vec{a}$ lies in xy -plane
 $|\vec{a}| = 18$ with 250° from x -ve
 $|\vec{b}| = 12$

$\vec{b}$ is in $+ve$ z -axis



$$\vec{c} = \vec{a} \times \vec{b}$$

$$= ab \sin \theta$$

$$= (18)(12) \sin 90$$

$$= 216$$

From right hand rule
we can say

$\vec{c}$ is in xy plane and $\perp$ to $\vec{a}$ (means 90° away from $\vec{a}$)

$$\theta_c = 250 - 90 = 160^\circ$$

Applied Physics

Chapter no : 4th

Motion in 2 & 3 D

Q no: 4

Given Data

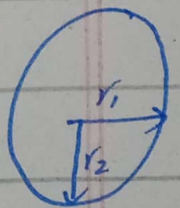
a) Magnitude = ?

Angle = ?

$$r_1 = 10\hat{i}$$

$$; \quad r_2 = -10\hat{j}$$

$$\Delta r = r_2 - r_1 = -10 - 10 = -10\hat{i} - 10\hat{j}$$



$$|\Delta r| = \sqrt{(10)^2 + (10)^2} = \sqrt{200} = 14.14$$

$$\tan \theta = \frac{-10}{-10} \Rightarrow \tan \theta = 1$$

$$\theta = 45^\circ$$

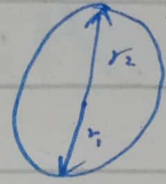
b) Given

$$|V| = ?$$

$$\theta = ?$$

$$r_1 = -10\hat{j} \quad ; \quad r_2 = 10\hat{j}$$

$$\Delta r = r_2 - r_1 = 10\hat{j} + 10\hat{j} = 20\hat{j}$$



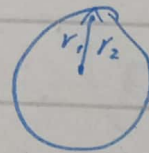
$$\tan \theta = \frac{20}{0} = \infty$$

$$\theta = 90^\circ$$

As final vector r_2 is in $+y$
so $\theta = 90^\circ$

c) $r_1 = 10\hat{j} \quad ; \quad r_2 = 10\hat{j}$
 $\Delta r = r_2 - r_1 = 0$

$$\theta = \tan^{-1} 0 = 0$$



Q no : 8

Given Data

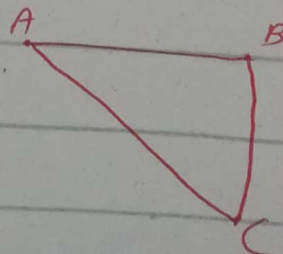
$$d_{AB} = 483 \text{ km}$$

$$d_{BC} = 966 \text{ km}$$

$$t_{AB} = 45 \text{ min}$$

$$t_{BC} = 1.5 \text{ h} = 90 \text{ min}$$

$$t = 45 + 90 = 135 \text{ min}$$



$$d_{AC} = d_{AB} + d_{BC}$$

$$= 483 \hat{i} - 966 \hat{j}$$

a) $|d_{AC}| = \sqrt{(483)^2 + (-966)^2}$

$$1.08 \times 10^3$$

b) $\theta = \tan^{-1} \frac{-966}{483} = -63.43^\circ$

$\theta = -63.43^\circ$ south of east
or 26.6° east of south

Average velocity

~~Avg Velocity = $\frac{|d_{AC}|}{t}$~~

~~$= \frac{483 \hat{i} - 966 \hat{j}}{145} \frac{1080 \text{ km}}{145 \text{ h}}$~~

~~$= 480 \text{ km/h}$~~

c) $t = 135 \text{ min}$ or $t = 2.25 \text{ h}$

$$V_{\text{Avg}} = \frac{|d_{AC}|}{t} = \frac{1080 \text{ km}}{2.25 \text{ h}}$$

$$= 480 \text{ km/h}$$

d) direction is same as in part (b)

Average Velocity

Assuming AB, BC a straight line so

$$\text{Speed}_{\text{avg}} = \frac{483 + 966}{2.25}$$

$$\text{Speed}_{\text{avg}} = 644 \text{ kmh}^{-1}$$

Q no: 9

From Graph

$$A(15, -15) \quad t_A = 0$$

$$B(30, -45) \quad t_B = 300$$

$$C(20, -15) \quad t_C = 600$$

$$D(45, 45) \quad t_D = 900$$

F //

$$\Delta r = r_2 - r_1$$

F //

$$V_{\text{avg}} = \frac{\Delta r}{\Delta t}$$

F //

$$\Delta r_{AB} = B - A = 15\hat{i} - 30\hat{j}$$

//

$$V_{AB} = \frac{\sqrt{(15)^2 - (30)^2}}{300} = 0.11 \text{ ms}^{-1}$$

$\vec{AC}$

$$r_{AC} = C - A = 5\hat{i} - 0\hat{j}$$
$$V_{AC} = \frac{\sqrt{(5)^2 - (0)^2}}{600} = 0.0083 \text{ ms}^{-1}$$

$\vec{AD}$

$$r_{AD} = D - A = 30\hat{i} + 60\hat{j}$$
$$V_{AD} = \frac{\sqrt{(30)^2 + (60)^2}}{900} = 0.07 \text{ ms}^{-1}$$

a) Avg velocity Magnitude

$$V_{AC} = 0.0083 \text{ ms}^{-1}$$

$$b) \theta = \tan^{-1}\left(\frac{-0}{5}\right) = 180^\circ$$

c) Avg velocity with greatest Magnitude

$$V_{AB} = 0.11 \text{ ms}^{-1}$$

d)

$$\theta = \tan^{-1} \frac{-30}{15} ; \theta = \tan^{-1}(2)$$

$$\theta = -63^\circ$$

Q no: 11

$$\vec{r} = (2t^3 - 5t)\hat{i} + (6 - 7t^4)\hat{j}$$

$$t = 2$$

$$v = ? \quad a = ? \quad r = ?$$

$$\begin{aligned} \text{a) } \vec{r} &= (2 \times 8 - 5 \times 2)\hat{i} - (6 - 7 \times 16)\hat{j} \\ r &= (6\hat{i} - 106\hat{j})\text{m} \end{aligned}$$

$$v = \frac{dr}{dt}$$

$$\begin{aligned} \text{b) } v &= (6t^2 - 5)\hat{i} + (-28t^3)\hat{j} \\ &= (6 \times 4 - 5)\hat{i} + (-28 \times 8)\hat{j} \\ &= (19\hat{i} - 224\hat{j})\text{ms}^{-1} \end{aligned}$$

$$\vec{a} = \frac{dv}{dt}$$

$$\begin{aligned} \text{c) } \vec{a} &= 12t\hat{i} - 84t^2\hat{j} \\ &= (24\hat{i} - 336\hat{j})\text{ms}^{-2} \end{aligned}$$

d) For angle of velocity

$$\theta = \tan^{-1} \left(\frac{-224}{19} \right)$$

$$\theta = -85.54^\circ$$

Q no: 14

$$\vec{v}_1 = 4\hat{i} - 2\hat{j} + 3\hat{k}$$

$$\vec{v}_2 = -2\hat{i} - 2\hat{j} + 5\hat{k}$$

$$\Delta t = 4 \text{ sec}$$

$$\begin{aligned} \text{a) } \vec{a}_{\text{avg}} &= \frac{\vec{v}_2 - \vec{v}_1}{\Delta t} = \frac{-6\hat{i} + 0\hat{j} + 2\hat{k}}{4} \quad // F \\ &= (-1.5\hat{i} + 0.5\hat{k}) \text{ ms}^{-2} \end{aligned}$$

$$\begin{aligned} \text{b) } |\vec{a}_{\text{avg}}| &= \sqrt{(-1.5)^2 + (0.5)^2} \\ &= 1.58 \end{aligned}$$

$$\text{c) } \theta = \tan^{-1} \left(\frac{y}{x} \right) = \tan^{-1} \left(\frac{0.5}{-1.5} \right)$$

$$\theta = -18^\circ$$

Q no: 18

$$\vec{a} = (5\hat{i} + 7\hat{j}) \text{ ms}^{-2}$$

$$\vec{v}_1 = 4\hat{i} \text{ ms}^{-1} \text{ at } t=0$$

As we know

$$\vec{a} = \vec{v}_2 - \vec{v}_1$$

$$\vec{v}_2 = \vec{a} + \vec{v}_1$$

$$\vec{v}_2 = 9\hat{i} + 7\hat{j}$$

Q no: 18

$$\vec{a} = (5\hat{i} + 7\hat{j}) \text{ ms}^{-2}$$

At $t=0$

$$v = (4\hat{i}) \text{ ms}^{-1}$$

$$v_{ix} = 4 \text{ ms}^{-1}$$

$$a_x = 5 \text{ ms}^{-2} \quad ; \quad a_y = 7 \text{ ms}^{-2}$$

$$x_2 - x_1 = 12 \text{ m}$$

As we know

$$F // \quad S = v_i t + \frac{1}{2} a t^2$$

$$F // \quad x_2 - x_1 = v_{ix} t + \frac{1}{2} a_x t^2$$

$$12 = 4t + \frac{1}{2} 5 t^2$$

$$24 = 8t + 5t^2$$

$$5t^2 + 8t - 24 = 0$$

Applying quadratic formula

$$t = 1.53 \text{ sec.}$$

Now for x and y coordinates of v_f .

$$\begin{aligned} F // \quad v_{fx} &= v_{ix} + a_x t \\ &= 4 + 5(1.53) \\ &= 11.65 \text{ ms}^{-1} \end{aligned}$$

$$\begin{aligned}
 V_{fy} &= V_{iy} + a_y t \\
 &= 0 + 7(1.53) \\
 &= 10.71 \text{ ms}^{-1}
 \end{aligned}$$

// F

$$a) \quad V_f = 11.65 \hat{i} + 10.71 \hat{j}$$

$$\begin{aligned}
 |V_f| &= \sqrt{(11.65)^2 + (10.71)^2} \\
 &= 15.82
 \end{aligned}$$

$$b) \quad \theta = \tan^{-1} \left(\frac{10.71}{11.65} \right)$$

$$\theta = 42.59^\circ$$

Q no: 28

$$V_i = 42 \text{ ms}^{-1}$$

$$\theta = 60^\circ$$

$$\text{time taken} = 5.5 \text{ s}$$

$$h = ? ; H = ? ; V_f = ?$$

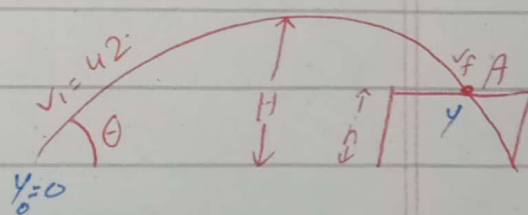
$$S = v_i t + \frac{1}{2} a t^2$$

$$y - y_0 = v_{iy} \sin \theta t - \frac{1}{2} g t^2 \quad \left(\begin{array}{l} \text{(-ve is because} \\ \text{of } g) \end{array} \right)$$

// F

$$y = y_0 + v_{iy} \sin \theta t - \frac{1}{2} g t^2$$

$$y = h \quad \therefore y_0 = 0$$



$$F // a) \quad y = y_0 + v \sin \theta t - \frac{1}{2} g t^2$$

$$y = 0 + 42 \sin 60 (5.5) - \frac{1}{2} (9.8) (5.5)^2$$

$$y = 200 - 148.225$$

$$y = 51.8 \text{ m}$$

$$F // b) \quad v_{fy} = v \sin \theta - g t$$

$$= 42 \sin 60 - (9.8) (5.5)$$

$$= -17.5 \text{ ms}^{-1}$$

Before hitting A speed of stone would be -17.5 .

// So, v becomes

$$F // \quad v_x = v \cos \theta = 42 \cos 60 = 21 \text{ ms}^{-1}$$

$$// \quad \vec{v} = \cancel{21 \text{ ms}^{-1}} 21 \text{ ms}^{-1} \hat{i} - 17.5 \text{ ms}^{-1} \hat{j}$$

$$F // c) \quad H = \frac{v^2 \sin^2 \theta}{2g}$$

$$H = \frac{(42)^2 (\sin 60)^2}{2(9.8)}$$

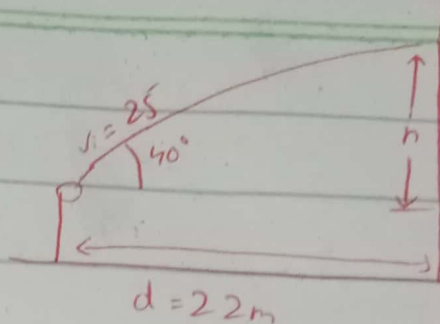
$$H = 67.5 \text{ m}$$

Q no: 32

$$v_i = 25 \text{ ms}^{-1}$$

$$\theta = 40^\circ$$

$$d = 22 \text{ m}$$



Solution:

$$v_{ix} = v_i \cos \theta = 25 \cos 40$$

$$v_{ix} = 19.15 \text{ ms}^{-1}$$

$$v_{iy} = v_i \sin \theta = 25 \sin 40$$

$$v_{iy} = 16.06 \text{ ms}^{-1}$$

To find time of flight

$$s = vt$$

$$t = \frac{d}{v_x} = \frac{22}{19.15} \quad \therefore s = d$$

$$t = 1.148 \text{ sec or } 1.15 \text{ sec}$$

$$a) \Delta y = v_{iy} t - \frac{1}{2} g t^2$$

$$\Delta y = 16.06 (1.15) - \frac{1}{2} (9.8) (1.15)^2$$

$$\Delta y = 12 \text{ m} \approx (11.988) \text{ m}$$

b)

$$V_x = V_i \cos \theta$$

$$V_x = 25 \cos 40$$

$$V_x = 19.2 \text{ ms}^{-1}$$

c)

$$V_{fy} = v_{iy} - gt$$

$$V_{fy} = 16.06 - (9.8)(1.15)$$

$$V_{fy} = 4.8 \text{ ms}^{-1}$$

d)

$$H = \frac{v_i^2 \sin^2 \theta}{2g}$$

$$H = \frac{(25)^2 (\sin 40)^2}{2(9.8)}$$

$$H = 13.17$$

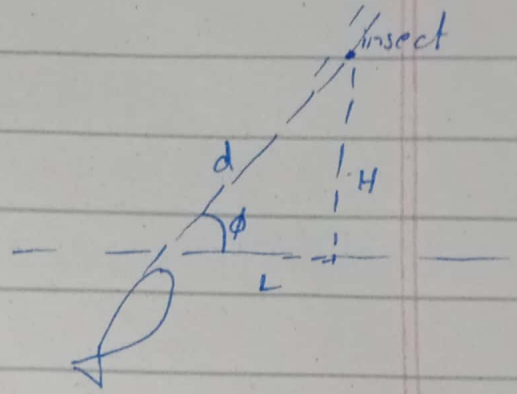
No, it has not passed highest point of trajectory which is 13.17m, it just hit on the height of 12m.

Q no: 41

$$H = d \sin \phi$$

$$L = d \cos \phi$$

$$\frac{H}{L} = \tan \phi$$



$$H = H_{\max} = \frac{v_i^2 \sin^2 \theta}{2g}$$

$$\text{Range} = L = \frac{L}{2} = \frac{v_i^2 \sin 2\theta}{2g} \Rightarrow \frac{2v_i^2 \sin \theta \cos \theta}{2g}$$

$$L = \frac{v_i^2 2 \sin \theta \cos \theta}{2g} \Rightarrow \frac{v_i^2 \sin \theta \cos \theta}{g}$$

$$\frac{H}{L} = \tan \phi$$

$$\frac{\frac{v_i^2 \sin^2 \theta}{2g}}{\frac{v_i^2 \sin \theta \cos \theta}{g}} = \tan \phi$$

$$\frac{1}{2} \tan \theta = \tan \phi$$

$$\tan \theta = 2 \tan \phi$$

$$\theta = \tan^{-1}(2 \tan \phi)$$

$$\theta = 55.46^\circ$$

Q no: 42

$$v_i = 26.5 \text{ ms}^{-1}$$

$$\theta = 53^\circ$$

$$H = ?$$

$$R = ?$$

$$\text{1st wheel height } y = ?$$

a) $s = vt$ $\therefore s = 23$ given

$$t = \frac{s}{v} = \frac{23\text{m}}{v_i \cos 53} = \frac{23}{15.44}$$

$$t = 1.44 \text{ sec}$$

$$y = y_0 + v_i \sin \theta t - \frac{1}{2} g t^2$$

$$Y = 3 + 26.5 \sin(53)(1.44) - \frac{1}{2}(9.8)(1.44)^2$$

$$Y = 3 + (30.47 - 10.16)$$

$$Y = 23.33 \text{ m}$$

Height of wheel is 18m
So

$$H_1 = Y - H_{\text{wheel}}$$

$$\underline{\underline{23.33 - 18}}$$

$$= 5.33 \text{ m}$$

b) $H_2 = H_{\text{max}}$

$$H_{\text{max}} = H_0 + \frac{v_i^2 \sin^2 \theta}{2g}$$

$$H_{\text{max}} = 3 + \frac{(26.5)^2 (\sin 53)^2}{2(9.8)}$$

$$H_{\text{max}} = 3 + 22.85$$

$$H_{\text{max}} = 25.85$$

$$H_2 = H_{\text{max}} - H_{\text{wheel}}$$

$$= 25.85 - 18$$

$$= 7.85 \text{ m}$$

c) $\text{Range} = \frac{v_i^2 \sin 2\theta}{g}$

$$\text{Range} = \frac{(26.5)^2 \sin(2 \times 53)}{9.8}$$

$$= 68.88 \text{ m}$$

Q no: 43

$$v = (7.6 \hat{i} + 6.1 \hat{j}) \text{ ms}^{-1} \text{ at}$$

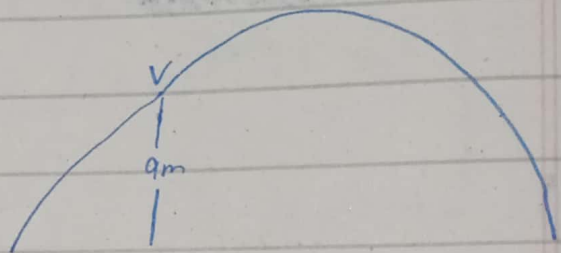
$$h_0 = 9 \text{ m}$$

$$H_{\text{max}} = ?$$

$$R_{\text{max}} = ?$$

$$|v_f| = ?$$

$$v_f \theta_f = ?$$



To find y-component of initial velocity

~~$$v_{fy}^2 = v_{iy}^2 - 2gS$$~~

$$2aS = v_{fy}^2 - v_{iy}^2$$

$$v_{iy}^2 = v_{fy}^2 - 2gS$$

$$v_{iy} = \sqrt{(6.1)^2 - 2(-9.8)(9)}$$

$$v_{iy} = \sqrt{213.61}$$

$$v_{iy} = 14.61 \text{ ms}^{-1}$$

At maximum height $v_f = 0$

~~$v_i = v_f$~~

a) $2gh = v_{fy}^2 - v_{iy}^2$

$$h = \frac{(0)^2 - (14.61)^2}{2(-9.8)}$$

$$h = 10.89 \approx 11\text{m}$$

Finding time

when ball comes to ground
 $y = 0$

$$y = v_{iy}t + \frac{1}{2}gt^2$$

$$0 = v_{iy}t + \frac{1}{2}gt^2$$

$$\frac{1}{2}gt^2 = -v_{iy}t$$

$$\frac{t^2}{t} = \frac{2v_{iy}}{g}$$

$$t = \frac{2v_{iy}}{g}$$

Finding horizontal distance

b) $x = v_{ix}t - \frac{1}{2}a_x t^2$

$$x = v_{ix}\left(\frac{2v_{iy}}{g}\right) - \frac{1}{2}(0)\left(\frac{2v_{iy}}{g}\right)^2$$

$$x = v_{ix} \left(\frac{2v_{iy}}{g} \right)$$

$$x = 7.6 \times 2 \frac{14.61}{9.8}$$

$$x = 22.66 \text{ m}$$

c) $v_{fx} = 7.6\hat{i}$ $v_{fy} = -14.61$

$$|v_f| = \sqrt{(7.6)^2 + (-14.61)^2}$$

$$|v_f| = \sqrt{271.21}$$

$$|v_f| = 16.46 \text{ ms}^{-1}$$

$$\theta = \tan^{-1} \frac{-14.61}{7.6}$$

$$\theta = -62.51^\circ$$

Q no: 53

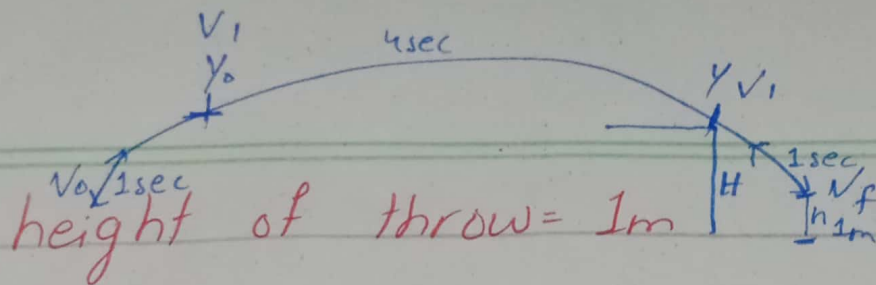
$$D = 50 \text{ m}$$

time from hand to top = 1 sec
top to hand

time on top = 4 sec

Total time = time on top + hand to top
+ top to hand

$$= 4 + 1 + 1 = 6 \text{ sec}$$



Solution.

$$S = vt$$

$$v_x = \frac{D}{t} = \frac{50}{4} = 12.5 \text{ ms}^{-1}$$

a)

$$S = vt$$

$$S = (12.5)(6)$$

$$S = 75.0 \text{ m}$$

$$V_{0x} = V_{1x} = V_{fx} = 12.5 \text{ ms}^{-1}$$

horizontal component

remain same.

As y_0 & y is at same position
so values will be same

b)

$$y - y_0 = v_{iy}t - \frac{1}{2}gt^2$$

$$0 = v_{iy}4 - \frac{1}{2}(9.8)(4)^2$$

$$v_{iy}4 = 78.4$$

$$v_{iy} = 19.6 \text{ ms}^{-1}$$

Now, to find v_{ix}

$$v_{iy} = v_{0y} - gt$$

$$19.6 = v_0 - (9.8)(1)$$

$$v_0 = 19.6 + 9.8$$

$$v_0 = 29.4 \text{ ms}^{-1}$$

$$V = (12.5 \hat{i} + 29.4 \hat{j}) \text{ ms}^{-1}$$

$$|V| = \sqrt{(12.5)^2 + (29.4)^2}$$

$$|V| = 31.94 \text{ ms}^{-1}$$

c) Height of Wall
Considering ~~1s~~ 1 sec of
motion in start

$$H - h = v_{iy} t - \frac{1}{2} g t^2$$

$$H = h + v_{iy} t - \frac{1}{2} g t^2$$

$$H = 1 + 29.4(1) - \frac{1}{2} (9.8)(1)^2$$

$$H = 30.4 - 4.9$$

$$H = 25.5 \text{ m}$$

Q no: 59

$$r = 15\text{m}$$

$$\omega = 5 \text{ rev/min} \quad f = 5 \text{ rev/min}$$

$$f = 5 \div 60 = 1/12 \text{ Hz/rev/sec}$$

$$a) \quad T = \frac{1}{f} = \frac{1}{1/12} = 12 \text{ sec}$$

Speed of ferris

$$V = \frac{2\pi r}{T} = \frac{2(3.14)(15)}{12}$$

$$V = 7.85 \text{ ms}^{-1}$$

$$b) \quad a = \frac{v^2}{r} = \frac{(7.85)^2}{15} = 4.11 \text{ ms}^{-2}$$

c) At highest point direction of centripetal acceleration is downward towards centre.

d) Magnitude will be same as in part (b)

e) The direction will be up towards centre of orbit.

Q no: 61

$$r = 20 \text{ km}$$

$$f = 1 \quad ; \quad T = \frac{1}{f} = 1$$

$$v = ? \quad a = ?$$

$$a) \quad v = \frac{2\pi r}{T} = \frac{2 \times (3.14) \times (2 \times 10^3)}{1}$$

$$v = 125663.7 \text{ ms}^{-1}$$

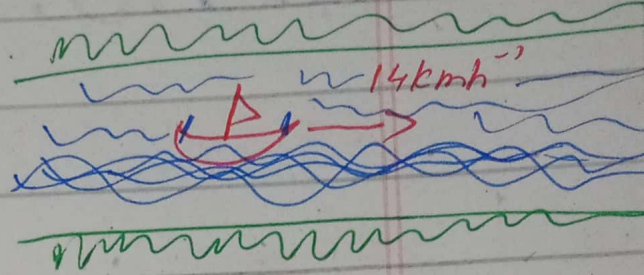
$$v = 1.25 \times 10^5 \text{ ms}^{-1}$$

$$b) \quad a = \frac{v^2}{r} = \frac{(1.25 \times 10^5)^2}{r}$$

$$a = 7.9 \times 10^5 \text{ ms}^{-2}$$

c) If star rotates faster then f increase, T decrease and ultimately both v and a will increase.

Q no: 70



$$V_{BW} = 14 \text{ kmh}^{-1}$$

$$V_{WG} = 9 \text{ kmh}^{-1}$$

Q

a)

$$\begin{aligned} V_{BG} &= V_{BW} - V_{WG} \\ &= 14 - 9 \\ &= 5 \text{ kmh}^{-1} \end{aligned}$$

Direction of V_{BG} is in +ve x-direction or upstream

b)

$$V_{CB} = 6 \text{ kmh}^{-1}$$

$$\begin{aligned} V_{CG} &= V_{CB} - V_{BG} \\ &= 6 - 5 \\ &= 1 \text{ kmh}^{-1} \end{aligned}$$

Direction of V_{CG} is in -ve x-direction or down stream.

A Aeroplane
W Wind
G Ground

Q no: 704

$$V_{\text{wind}} = 42 \text{ kmh}^{-1}$$

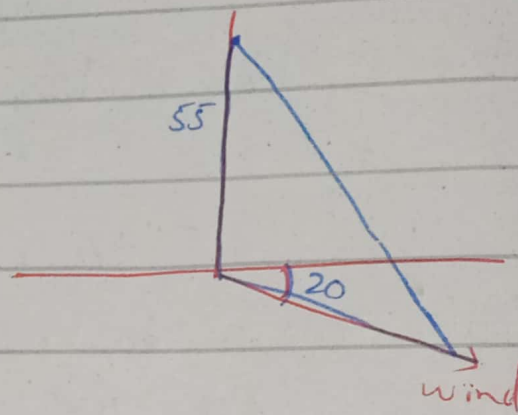
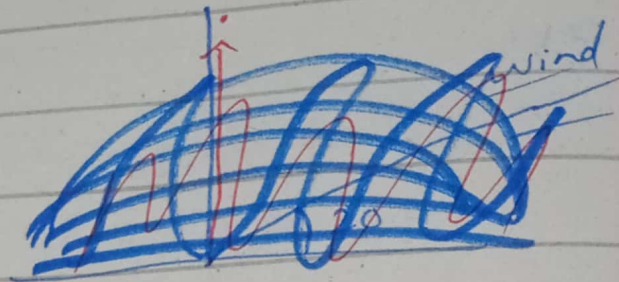
~~V_{airplane}~~ =

$$t = 15 \text{ min.}$$

$$= 0.25 \text{ h}$$

$$V_{\text{AW}} = ?$$

$$d_{\text{Airplane}} = 55 \text{ km}$$



Solution

$$V_{\text{AG}} = \frac{S}{t} = \frac{55 \text{ km}}{0.25} = 220 \hat{j} \text{ kmh}^{-1}$$

$$V_{\text{AG}} = 220 \text{ kmh}^{-1}$$

$$V_{\text{WG}} = 42 \cos 20^\circ \hat{i} - 42 \sin 20^\circ \hat{j} \\ = (39.5 \hat{i} - 14.36 \hat{j}) \text{ kmh}^{-1}$$

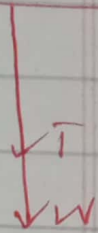
$$V_{\text{AW}} = V_{\text{AG}} - V_{\text{WG}} \\ = 220 \hat{j} - 39.5 \hat{i} + 14.36 \hat{j} \\ = -39.5 \hat{i} + 234.36 \hat{j}$$

$$|V_{\text{AW}}| = \sqrt{(-39.5)^2 + (234.36)^2} \\ |V_{\text{AW}}| = 237.66 \text{ kmh}^{-1}$$

Q no: 75

$$V_{TG} = 30 \text{ ms}^{-1}$$

$$V_{RG} = ?$$



$$\theta_r = 70^\circ \text{ +ve y axis}$$

~~or 20° with x-axis~~

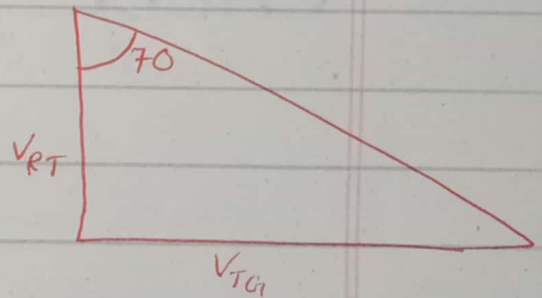
Q no: 75

For train

$$V_{TG} = 30 \text{ ms}^{-1}$$

$$V_{RG} = ?$$

$$\theta_{RG} = 70^\circ$$



$$\tan \theta = \frac{V_{TG}}{V_{RT}}$$

$$V_{RT} = \frac{V_{TG}}{\tan \theta} = \frac{30}{\tan 70}$$

$$V_{RT} = 10.91 \text{ ms}^{-1}$$

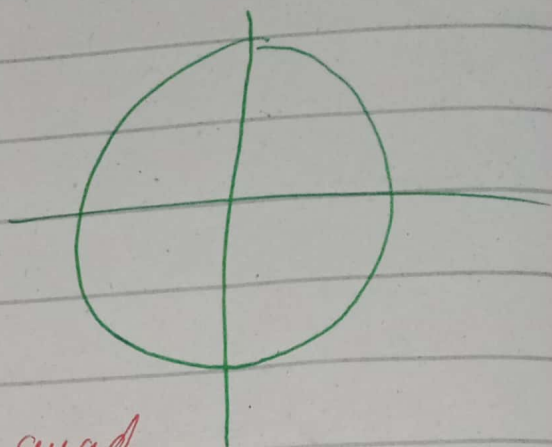
$$V_{RG} = V_{TG} \hat{i} + V_{RT} \hat{j}$$

$$|V_{RG}| = \sqrt{(30)^2 + (10.91)^2}$$

$$|V_{RG}| = 32 \text{ ms}^{-1}$$

Check-point 1

$$v = 2\hat{i} - 2\hat{j}$$



Clockwise = 1st quad

Counter-Clockwise = 3rd quad

Check-point 2

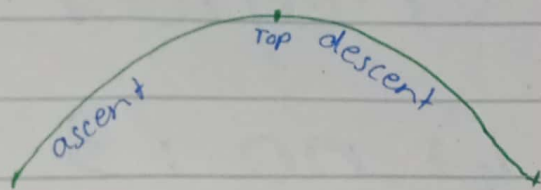
By taking derivatives of all points we can say acceleration is not constant.

Check-point 3

$$v = 25\hat{i} - 4.9\hat{j}$$

at highest point vertical component of velocity will be zero so it have not passed it highest point.

Check-point 4



- a) Horizontal component of velocity remain constant throughout the flight
- b) Vertical comp of v decreases in ascent, become zero on top and then increases in descent
- c) Horizontal comp of ~~velocity~~ acceleration remain zero.
- d) Vertical comp of acceleration is negative during ascent, zero on top and positive during descent with mag approx 9.8

Check-point - 5

at $x = -2\text{m}$ $v = -(4\text{ms}^{-1}) \cdot \hat{j}$

at $y = 2\text{m} = r$

a) velocity = $-(4\text{ms}^{-1}) \cdot \hat{j}$

b) acceleration = $\frac{v^2}{r} = \frac{16}{2} = +8\text{ms}^{-2}$ towards centre.

